Explicit irrationality bounds for odd powers of π

Nested logarithmic Hermite–Padé constructions,
prime divisors, and algebraic-approximation transfer

Expository research notes

Revised 31 October 2026

Abstract

We develop a nested-interval version of a classical logarithmic Hermite–Padé construction and give explicit upper bounds for the irrationality exponents of odd powers of π. The construction uses Gaussian-integer polynomials in $\log \mathrm{i} = \mathrm{i}\pi/2$, rather than a direct sequence of binary linear forms in $1, \pi^k$. We prove integrality, an exact nonvanishing determinant identity, a prime-divisor estimate described by weighted arcs on a circle, and uniform analytic bounds. A minimal-polynomial resultant argument handles every rational approximation, including reducible-binomial and rational-power cases. The latest twenty-eight-layer choice gives

$$\boxed{\mu(\pi^3) \le \frac{12970477737}{11706979} = 1107.9269670681\ldots < 1107.93.}$$

This fixed result, including the actual auxiliary family and the required arithmetic and analytic bounds, has an accepted Lean formalization. The general transfer applies to every integer power $k \ge 2$; further numerical instances and an explicit all-power recipe are explained with their separate verification status. We also derive an older all-power bound from Nesterenko–Waldschmidt and compare the even case with the specialized bound for $\zeta(4)$. Background, intermediate calculations, worked examples, and a guide to the formal proof are included.

Status and scope. This is an expanded working manuscript, not a journal submission or an independently refereed paper. It makes no priority, record, or global optimality claim. The displayed upper bounds follow from the arguments given here; finite tests and numerical certificates have more limited roles, which are stated explicitly. No upper bound below 100 for $\mu(\pi^3)$ is established. The exact L99 theorem is accepted locally; the entire expanded exposition and every additional example are not thereby claimed to be formalized.

Contents

1 Purpose, conventions, and main conclusions **3**
1.1 What is being bounded? . . . 3
1.2 Why odd powers are natural targets . . . 3
1.3 A preview of the construction . . . 4
1.4 General parameter theorem . . . 4
1.5 A guide to the evidence . . . 5

2 Background and a reading guide **6**
2.1 Irrationality, transcendence, and approximation are different questions . . . 6
2.2 What Padé approximation contributes . . . 6
2.3 Why evaluate at a complex point? . . . 6
2.4 The two parameters that must not be interchanged . . . 7
2.5 A map of the proof . . . 7

3 Prior explicit bounds and the even-power comparison **8**
3.1 An older theorem already gives a number for every power . . . 8
3.2 Even zeta values give an identification, not an approximation theorem . . . 9
3.3 What the same L99 shape would do for the fourth power . . . 9

4 The auxiliary polynomials and their determinant **10**
4.1 Intervals, poles, and normalization . . . 10
4.2 Why all coefficients are integers . . . 11
4.3 Supports and vanishing orders . . . 11
4.4 A nonzero determinant, including its leading term . . . 12

5 Prime divisors and weighted zero arcs **12**
5.1 A local valuation inequality . . . 12
5.2 Replacing pole locations by a circle . . . 13
5.3 The common divisor and its exact exponential rate . . . 14
5.4 A transparent two-layer special case . . . 15

6 Uniform analytic estimates **15**
6.1 A coefficient bound valid on a whole disk . . . 15
6.2 A global horizontal-contour bound . . . 16
6.3 Evaluating the central exponential rate . . . 16

7 From logarithmic polynomials to integer powers **17**
7.1 Why a resultant is the right arithmetic object . . . 17
7.2 The quantitative comparison . . . 18
7.3 Why a common divisor helps twice . . . 19

8 Concrete parameters and every fixed integer power **19**
8.1 Recovering the balanced benchmark . . . 19
8.2 Two, seven, and twelve layers . . . 19
8.3 The accepted L99 instance, step by step . . . 20
8.4 An explicit scale recipe for every fixed power . . . 21
8.5 Why the higher-power constants still grow rapidly . . . 22
8.6 A simple all-power bound from the L99 shape . . . 23
8.7 Sensitivity of the bound and safe numerical substitution . . . 24
8.8 What would be needed to reach 10–100? . . . 24

9 How the constants are computed and certified **24**
9.1 An exact finite combinatorial calculation . . . 24
9.2 Ball arithmetic, root signs, and strict rounding . . . 25
9.3 Reproduction and the role of each file . . . 26
9.4 Parameter order and effectivity . . . 26

10 What has been formalized, and how the proof is organized **27**
10.1 The exact accepted statement . . . 27
10.2 From an abstract transfer to the actual family . . . 27
10.3 Why a midpoint table needs a real-variable proof . . . 28
10.4 How finite windows prove an infinite asymptotic estimate . . . 28
10.5 Why the numerical proof does not trust floating point . . . 28
10.6 Acceptance evidence and its limits . . . 29

A Small examples and useful background **30**
A.1 A complete two-by-two system . . . 30
A.2 Why the minimal polynomial cannot be skipped . . . 30
A.3 Divisibility before and after evaluation . . . 30
A.4 The least common multiple as an arithmetic cost . . . 30
A.5 A two-layer arc calculation by hand . . . 31

B Finite experiments and further directions **31**
B.1 Scalar Padé approximation to a pure cubic target . . . 31
B.2 A derivative kernel isolating $\beta(3)$ . . . 31
B.3 What a successful binary family would have to prove . . . 32
B.4 Corrections that should not become unproved claims . . . 32

C Notation, intermediate checks, and reading exercises **33**
C.1 A notation sheet . . . 33
C.2 Deriving the exponent without suppressing the index choice . . . 33
C.3 Short exercises with solutions or hints . . . 34

1 Purpose, conventions, and main conclusions

1.1 What is being bounded?

For an irrational real number ξ, its irrationality exponent is

$$\mu(\xi) = \sup\left\{u > 0 : 0 < \left|\xi - \frac{p}{q}\right| < q^{-u} \text{ for infinitely many reduced } p/q,\ q > 0\right\}.$$

Thus $\mu(\xi) \le E$ means that for every $\varepsilon > 0$ there are only finitely many reduced fractions with error below $q^{-E-\varepsilon}$. Equivalently, after changing a positive constant, one has

$$\left|\xi - \frac{p}{q}\right| \ge c_\varepsilon q^{-E-\varepsilon} \tag{1.1}$$

for all p, q with $q > 0$. This does not assert the same inequality with exponent exactly E.

Dirichlet's theorem gives $\mu(\xi) \ge 2$. Obtaining an explicit upper bound for a particular transcendental number is a separate problem. The finiteness of the exponents of fixed powers of π is already accessible through classical quantitative transcendence theory. Our objective is to spell out one concrete construction and track its constants, not to claim a first finiteness theorem.

All logarithms in estimates are natural logarithms. We use

$$\lambda = \log \mathrm{i} = \frac{\mathrm{i}\pi}{2}, \qquad \Theta = \frac{\pi}{2}.$$

The symbols $o(n)$ and $O(\log n)$ refer to $n \to \infty$ with the complete interval configuration and target k fixed. The auxiliary degree m may be large, but is fixed in this limit.

1.2 Why odd powers are natural targets

The Dirichlet beta function is

$$\beta(s) = \sum_{\ell=0}^{\infty} \frac{(-1)^\ell}{(2\ell+1)^s}.$$

For $r \ge 0$,

$$\beta(2r+1) = \frac{(-1)^r E_{2r}}{4^{r+1}(2r)!}\pi^{2r+1},$$

where $\operatorname{sech} z = \sum_{j\ge 0} E_j z^j/j!$ defines the Euler numbers. In particular,

$$\beta(1) = \frac{\pi}{4}, \qquad \beta(3) = \frac{\pi^3}{32}, \qquad \beta(5) = \frac{5\pi^5}{1536}.$$

Multiplication by a nonzero rational number preserves the irrationality exponent. For a reduced multiplier a/b, reducing $ap/(bq)$ cancels a gcd dividing $|ab|$, so its denominator remains comparable to q by constants independent of p, q.

This observation does *not* deduce an upper bound for π^3 from a rational-approximation bound for π. Taking a cube root changes the approximation field. The resultant argument below accounts for that change.

1.3 A preview of the construction

There are three components:

1. Build integer polynomials $A_{h,j,n}(x)$ such that
$$R_{h,n}(x) = \sum_{j=0}^{m} A_{h,j,n}(x)(\log x)^j$$
has a high-order zero at $x = 1$.
2. Divide every coefficient by a common integer Π_n. Its exponential rate $\gamma = \lim n^{-1} \log \Pi_n$ matters; a large constant depending only on m does not change this rate.
3. Evaluate at $x = \mathrm{i}$, compare λ with a k-th root of a rational approximation to π^k, and use a nonzero resultant.

Nested intervals change both analytic bounds and arithmetic divisors. Their effects must be assessed together.

The classical logarithmic construction appears in Mignotte [1]. Weighted logarithmic Padé systems and normality are discussed in Kawashima [2]. Here the determinant and estimates for our two-sided nested system are proved directly.

1.4 General parameter theorem

Choose integers
$$L \geq 3, \quad 0 = t_0 < t_1 < \cdots < t_{J-1} < L/2, \quad \nu_j \geq 1,$$
and put
$$M = \sum_{j=0}^{J-1} \nu_j, \qquad m = M - 1 \geq 1, \qquad c_j = 1 - \frac{2t_j}{L}.$$
An interval of relative length c_j, centred at $1/2$, is used ν_j times. Let γ be the explicit prime-divisor rate defined in Section 5. Let $Y > 0$ solve
$$2 \sum_j \nu_j \arctan \frac{c_j}{2Y} = \Theta. \tag{1.2}$$
Define
$$F_c(Y) = \frac{c}{2} \log \frac{1 + (2Y/c)^2}{4} + 2Y \arctan \frac{c}{2Y}, \tag{1.3}$$
$$D_0 = \sum_j \nu_j F_{c_j}(Y) - \Theta Y - m, \tag{1.4}$$
$$G_0 = m + \left(\sum_j \nu_j c_j \right) \log 2, \tag{1.5}$$
$$D = D_0 + \gamma, \qquad G = G_0 - \gamma. \tag{1.6}$$

Theorem 1.1 (Integer-power transfer bound)**.** *Let $k \geq 2$ be an integer. If $G \geq 0$ and $\Delta_k = D - (k-1)G > 0$, then*
$$\boxed{\mu(\pi^k) \leq E_k := \frac{m(D+G)}{D-(k-1)G} = \frac{m(D_0+G_0)}{D_0-(k-1)G_0+k\gamma}.} \tag{1.7}$$

Odd powers are the motivating targets, but no parity hypothesis is needed in the theorem or its accepted general Lean transfer. The specialized even-power literature is discussed in Section 3. An instance needs its own positive gap; the cubic parameters cannot simply be reused for every k.

Corollary 1.2 (Accepted L99 cubic instance)**.** *Take $L = 99$, $t_j = j$ for $0 \le j \le 27$, and*

$$\nu_0 = 64, \qquad \nu_j = 4 \ (1 \le j \le 27), \qquad m = 171.$$

The actual family admits the conservative rational rates

$$D = \frac{54469491}{100000}, \qquad G = \frac{2672657}{12500}, \qquad D - 2G = \frac{11706979}{100000} > 0.$$

Consequently

$$\mu(\pi^3) \le \frac{12970477737}{11706979} < 1107.93.$$

The fixed assertion, with these exact rational constants, is accepted in Lean. Its construction and arithmetic ledger are given in Section 8.3; the meaning and boundaries of formal acceptance are explained in Section 10.

Corollary 1.3 (A twelve-layer family)**.** *Use $L = 69$, $t_j = j$ for $0 \le j \le 11$, and*

$$\nu_0 = 6s, \qquad \nu_1 = \cdots = \nu_{11} = s, \qquad m = 17s - 1.$$

The following choices satisfy the hypotheses:

k	s	m	*Certified strict upper bound*
3	*11*	*186*	1203.973
5	*259*	4 402	42 461
7	5 871	99 806	1 272 103
9	131 354	2 233 017	35 398 937

For every fixed integer $k \ge 2$, some sufficiently large integer s also satisfies the hypotheses.

The cubic entry is retained as a historical twelve-layer comparison; Corollary 1.2 is now the stronger accepted cubic result. The other table entries have manuscript proofs and Arb certificates, but are not accepted fixed Lean applications. A definite scale recipe and a deliberately coarse bound $301 \cdot 8^k$ are proved on paper in Section 8. Neither replaces the specialized even-power literature nor improves the classical all-power estimate for large k.

1.5 A guide to the evidence

The algebraic and analytic lemmas prove Theorem 1.1. Ball-arithmetic certificates prove the decimal comparisons once the lemmas hold. Exact finite examples check identities and implementation, but do not prove asymptotic rates. Floating-point searches select parameters; they neither certify rounding nor prove global optimality. The appendices retain unsuccessful experiments because they explain the arithmetic difficulties that remain.

There are three different objects to keep in view: a mathematical argument for a family of estimates, finite certificates for specified real-number comparisons, and a checked Lean theorem with an exact dependency closure. For the accepted L99 instance these fit together. For the higher-power examples only the first two have been assembled. This distinction is part of the statement of the results, rather than an implementation detail to be suppressed.

2 Background and a reading guide

2.1 Irrationality, transcendence, and approximation are different questions

An irrationality proof rules out an equality $\xi = p/q$. An irrationality-exponent bound rules out an unlimited supply of exceptionally accurate near-equalities. Transcendence rules out all polynomial equalities with integer coefficients, but by itself does not quantify how small a nonzero polynomial value may be. The last distinction is essential here: the transcendence of π already implies that π^k is irrational for every positive integer k, yet it does not supply the numerical exponent in (1.1).

For example, the statement $\mu(\xi) \le E$ controls infinitely many denominator scales. A long continued-fraction computation supplies finitely many very good approximations, and may reveal useful patterns, but cannot exclude a new pattern at larger denominators. The present proof therefore constructs auxiliary objects for every sufficiently large index n, and relates that index to an arbitrary denominator q. This is the reason for the uniformity requirements in Sections 6 and 7.

It is also useful to separate three quantities often called a "measure." A numerical exponent E specifies the power of q. A positive prefactor c_ε specifies the scale of the lower bound. A denominator threshold $q_0(\varepsilon)$ specifies where that lower bound starts. These notes make E explicit, but do not tabulate numerical values for the other two quantities in the nested construction. Allowing an arbitrary positive exponent slack is part of the theorem, not a rounding convention.

2.2 What Padé approximation contributes

For a power series $f(x)$, ordinary Padé approximation seeks polynomials P, Q for which $Q(x)f(x) - P(x)$ has a prescribed high-order zero. Hermite–Padé approximation replaces a single function by several functions. In our setting these are

$$1, \log x, (\log x)^2, \dots, (\log x)^m.$$

For each row h, the expression

$$R_{h,n}(x) = A_{h,0,n}(x) + A_{h,1,n}(x)\log x + \cdots + A_{h,m,n}(x)(\log x)^m$$

vanishes to high order at $x = 1$. At a fixed evaluation point this can make its value small even though its coefficients are large. The integral representation is an efficient way to impose many vanishing conditions simultaneously.

The vanishing conditions are linear equations, but solving a large rational linear system independently at every n would conceal the denominators and make nonvanishing difficult to track. The explicit rational kernel instead exposes its poles, factorials, and local expansions. These are exactly the objects needed later for integrality and prime-divisor estimates.

There are $m+1$ rows because a single small polynomial may vanish at the algebraic number selected by a rational approximation. An invertible coefficient matrix ensures that the entire family cannot vanish there. The exact determinant is therefore an arithmetic selection tool: it supplies at least one useful row at every denominator scale.

2.3 Why evaluate at a complex point?

The principal logarithm at i is $\lambda = \mathrm{i}\pi/2$, so powers of the logarithm encode powers of π. An integer polynomial evaluated at i has coefficients in $\mathbb{Z}[\mathrm{i}]$, a ring whose nonzero elements have

modulus at least one. This discreteness replaces the familiar lower bound $|a| \geq 1$ for a nonzero ordinary integer.

No irrationality conclusion follows merely from making $U_{h,n}(\lambda)$ small: its degree is usually much larger than the target power k. The next step is to compare it with $U_{h,n}(y)$, where

$$y = \frac{\mathrm{i}}{2}(p/q)^{1/k}$$

is algebraic. The comparison is useful because an algebraic value has a nonzero integral resultant, while the logarithmic value is small by analysis. The divided difference controls the cost of moving from λ to y.

For odd powers, an alternative motivation comes from the rational multiple $\beta(2r+1)/\pi^{2r+1}$. That identity suggests trying binary linear forms in $1, \beta(3)$, but it does not require us to find them. The present method uses a higher-dimensional system and a resultant instead. The unsuccessful binary experiments in Appendix B explain why this detour was useful.

2.4 The two parameters that must not be interchanged

The interval configuration, its multiplicities, and m are fixed before the limit $n \to \infty$. The index n, taken through multiples of L, determines the number of poles in the rational kernel. The degree m determines the number of logarithmic powers and rows. They have different roles even when both are large.

For instance, in the accepted L99 application $m = 171$, whereas n is unbounded. A factor $171!$ is a constant with respect to n, so its logarithm divided by n tends to zero. A factor d_n^{171} is not a constant: its exponential rate is 171. Treating these two factors alike can easily reverse the apparent success of a construction.

When the target power k changes, we may select another fixed scale s, hence another fixed m. The all-power theorem means that this can be done for each k, followed by the separate limit in n. It does not authorize a simultaneous limit $k = k(n)$ in the estimates already proved.

2.5 A map of the proof

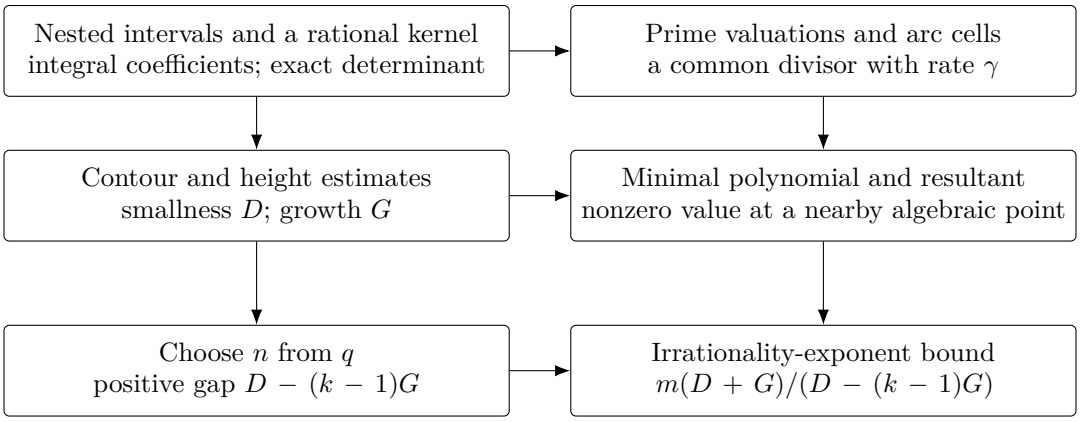


Figure 1: The proof separates construction, arithmetic, analysis, and transfer. A numerical improvement must respect all four parts.

A first reading can follow the overview, the L99 ledger in Section 8.3, and the transfer proof. A second reading can fill in the local coefficient expansion and the contour tails. The arc

partition and the formalization guide are best read after the purpose of a common divisor is clear. Appendix C collects notation and short checks that can be carried out by hand.

The main prerequisites are elementary complex analysis, polynomial algebra over $\mathbb{Q}$ and $\mathbb{Q}(\mathrm{i})$, prime valuations, and asymptotic estimates. Resultants, confluent Vandermonde matrices, and interval arithmetic are introduced where they enter the proof. Knowledge of Lean is not required to read the mathematical argument.

3 Prior explicit bounds and the even-power comparison

3.1 An older theorem already gives a number for every power

The distinction between an explicitly tabulated bound and a bound that follows by substitution into a general theorem matters here. Nesterenko–Waldschmidt [4], Theorem 2(1), prove that if ξ is real algebraic of degree d and coefficient length at most $\mathcal{L} \geq 3$, then

$$|\pi - \xi| \geq \exp\{-1200000d(\log \mathcal{L} + d \log d)(1 + \log d)\}. \tag{3.1}$$

The coefficient length of an integral polynomial is the sum of the absolute values of its coefficients; for an algebraic number it is the length of its primitive integral minimal polynomial. The following specialization is derived here from their theorem.

Proposition 3.1 (A classical all-power consequence)**.** *For every integer $k \geq 1$, put*

$$B(k) = 1200000k(1 + \log k).$$

Then $\mu(\pi^k) \leq B(k)$. In particular, explicit exponent upper bounds for all fixed powers were already available from classical quantitative transcendence theory.

Proof. It suffices first to consider $q \geq 1$ and $|\pi^k - p/q| < 1$. This implies $p/q > 0$. Set $\theta = (p/q)^{1/k}$, and let f be its primitive integral minimal polynomial, of degree $d \leq k$ and positive leading coefficient a. Gauss's lemma gives $f \mid qX^k - p$ in $\mathbb{Z}[X]$, after removing the content of the latter polynomial if necessary. In particular $a \leq q$. Every conjugate of θ is a root of this binomial, hence has modulus θ. Also $\theta < \pi + 1$. The elementary bound for the coefficients of a product gives

$$\mathcal{L}(f) \leq a(1 + \theta)^d < 6^k q.$$

This step uses the minimal polynomial and remains valid when the binomial is reducible.

Use $\mathcal{L} = 6^k q$ in (3.1). Monotonicity in $1 \leq d \leq k$ yields

$$|\pi - \theta| \geq \exp\{-B(k)(\log q + k \log 6 + k \log k)\} = (6k)^{-kB(k)} q^{-B(k)}.$$

Finally,

$$|\pi^k - p/q| = |\pi - \theta| \sum_{a=0}^{k-1} \pi^{k-1-a}\theta^a \geq \pi^{k-1}|\pi - \theta|.$$

For the fractions excluded at the start the error is at least one. Thus the smaller prefactor $(6k)^{-kB(k)}$ gives a lower bound for all integer p and $q \geq 1$, proving the exponent claim. □

This consequence is not a claim that the original paper prints a table of π^k exponents. It shows why absence of such a table would not establish priority for an explicit number. It also gives an independent comparison whose growth is only $O(k \log k)$.

k	Classical consequence $B(k)$, approximately	Bound displayed in these notes
3	7 555 004.24	1107.93
5	15 656 627.47	42 461
7	24 745 645.25	1 272 103
9	34 530 025.44	35 398 937

The first three entries show a substantial improvement over this particular classical consequence. The fourth does not: the old general theorem already beats the displayed twelve-layer $k = 9$ number. None of these comparisons is an exhaustive literature search for the best bound at the indicated power. They delimit the contribution that can safely be claimed.

3.2 Even zeta values give an identification, not an approximation theorem

Euler's formula, with B_{2r} the Bernoulli numbers, reads

$$\zeta(2r) = \frac{(-1)^{r+1} B_{2r} (2\pi)^{2r}}{2(2r)!} \in \mathbb{Q}^{\times} \pi^{2r}.$$

Therefore $\mu(\zeta(2r)) = \mu(\pi^{2r})$. For example,

$$\zeta(2) = \pi^2/6, \qquad \zeta(4) = \pi^4/90.$$

This equality transfers an existing irrationality-exponent estimate between two names for the same rationally scaled constant. It does not turn evaluation of the zeta value into a proof of a sharp irrationality measure. Nor does an upper bound for $\mu(\pi^2)$ automatically give the corresponding upper bound for $\mu(\pi^4)$. Approximating $(\pi^2)^2$ and approximating π^2 by rational numbers are different approximation problems.

Marcovecchio–Zudilin [5] establish the unconditional specialized estimate

$$\mu(\pi^4) = \mu(\zeta(4)) \leq 12.51085940\ldots. \tag{3.2}$$

Their 2019 preprint, published in 2020, uses a different hypergeometric representation to control the arithmetic of approximations to $\zeta(4)$. The bound does not assume the older denominator conjecture discussed in their introduction. It is the specialized benchmark verified for this revision; the present notes do not claim a complete current classification of all later work.

3.3 What the same L99 shape would do for the fourth power

For the accepted cubic rates, direct substitution gives

$$D - 3G = 544.69491 - 3(213.81256) = -96.74277.$$

Thus the fixed accepted cubic instance is inadmissible for $k = 4$. This is a failed hypothesis, rather than a failure of the general transfer theorem. Increasing the common multiplicity scale changes both the degree and the decay-to-height ratio.

For orientation, keep the base L99 shape $(16, 1, \ldots, 1)$, and use the root-free lower rate of Section 8.6 with scale $s = 19$. Then

$$m = 816, \qquad \underline{D}_s \approx 3630.47940,$$
$$G_s \approx 1019.35966, \qquad \underline{D}_s - 3G_s \approx 572.40043.$$

The associated expression is approximately 6628.69640. The accompanying numerical check certifies that it is below 6628.70, using the already proved rational divisor lower rate. This is a manuscript-level specialization and an Arb comparison, not an accepted fixed Lean theorem for π^4. Its size is already sufficient to explain the comparison: the uniform method is far from (3.2).

Consequently there are two different useful projects. Completing general-power applications tests and documents a reusable proof framework. Improving the specialized fourth-power bound requires competing below $12.51085940\ldots$, most naturally by sharpening the arithmetic of direct $\zeta(4)$ approximations. Extending the range of a theorem and improving the best numerical constant are distinct achievements.

4 The auxiliary polynomials and their determinant

4.1 Intervals, poles, and normalization

Fix the parameters of Theorem 1.1. Let n run through positive multiples of L. List all $M = m + 1$ intervals, including repetitions, in nested order:

$$I_a = [\alpha_a, \beta_a], \quad 0 \le a \le m, \qquad 0 = \alpha_0 \le \cdots \le \alpha_m < n/2, \quad \beta_a = n - \alpha_a.$$

Thus $I_m \subseteq \cdots \subseteq I_0 = [0, n]$. Write $\ell_a = \beta_a - \alpha_a$. The grouped description uses endpoints $t_j n/L, (L - t_j)n/L$, with multiplicity ν_j.

Set

$$\begin{aligned} Q_n(z) &= \prod_{a=0}^{m} \prod_{r=\alpha_a}^{\beta_a} (z + r), \\ N_n &= \sum_{a=0}^{m} \ell_a, \qquad d_n = \operatorname{lcm}(1, \ldots, n), \\ C_n &= m!\, d_n^m \prod_{a=0}^{m} \ell_a!. \end{aligned} \tag{4.1}$$

The degree of Q_n is $N_n + M$. *The interval factorials are in the numerator of C_n.*

Let Γ positively surround the poles $0, -1, \ldots, -n$. On a local logarithm branch define

$$R_{h,n}(x) = \frac{C_n}{2\pi \mathrm{i}} \oint_\Gamma \frac{z^h x^{z+n}}{Q_n(z)}\, \mathrm{d}z, \qquad 0 \le h \le m. \tag{4.2}$$

Write

$$\frac{z^h}{Q_n(z)} = \sum_{\ell=0}^{n} \sum_{r=1}^{\mu_\ell} \frac{b_{h,\ell,r}}{(z + \ell)^r}, \quad \mu_\ell = \#\{a : \ell \in I_a\}.$$

Residues give

$$R_{h,n}(x) = \sum_{j=0}^{m} A_{h,j,n}(x)(\log x)^j, \tag{4.3}$$

$$A_{h,j,n}(x) = \frac{C_n}{j!} \sum_{\ell : \mu_\ell \ge j+1} b_{h,\ell,j+1} x^{n-\ell}. \tag{4.4}$$

4.2 Why all coefficients are integers

Lemma 4.1 (Integrality)**.** *For all h, j, one has $A_{h,j,n}(x) \in \mathbb{Z}[x]$.*

Proof. Absorb the interval factorials first:

$$H_a(z) = \frac{\ell_a!}{\prod_{r=\alpha_a}^{\beta_a}(z+r)}.$$

If $\ell \in I_a$, expansion at $z = -\ell + w$ gives

$$wH_a(-\ell+w) = (-1)^{\ell-\alpha_a}\binom{\ell_a}{\ell-\alpha_a}\prod_{\substack{r\in I_a\\ r\neq\ell}}\left(1+\frac{w}{r-\ell}\right)^{-1}. \tag{4.5}$$

The coefficient of w^u has denominator dividing d_n^u: every distance is a nonzero integer of absolute value at most n.

If $\ell \notin I_a$, use

$$H_a(z) = \sum_{r=\alpha_a}^{\beta_a}\frac{(-1)^{r-\alpha_a}\binom{\ell_a}{r-\alpha_a}}{z+r}. \tag{4.6}$$

Now the coefficient of w^u has denominator dividing d_n^{u+1}. The extra factor matters: a regular factor need not have an integral constant term.

There are μ_ℓ pole factors and $M - \mu_\ell$ regular factors. The coefficient of $(z+\ell)^{-j-1}$ requires total local order $\mu_\ell - 1 - j$. The polynomial $(-\ell+w)^h$ has integer coefficients and can only reduce the order required from the other factors. Thus the necessary exponent of d_n is at most

$$(\mu_\ell - 1 - j) + (M - \mu_\ell) = m - j.$$

Multiplication by $m! d_n^m / j!$ clears the denominators. □

The cancellation of μ_ℓ is why a variable pole multiplicity does not require a larger universal power than d_n^m.

4.3 Supports and vanishing orders

Nestedness implies

$$\mu_\ell \geq j+1 \quad\Longleftrightarrow\quad \ell \in [\alpha_j, \beta_j].$$

Hence $A_{h,j,n}$ is divisible by $x^{n-\beta_j}$, and the remaining degree is at most ℓ_j.

Lemma 4.2 (Vanishing at 1)**.** *Every $R_{h,n}$ has a zero of order at least N_n at $x = 1$.*

Proof. Put $w = \log x$. The coefficient of w^r in the integral of $z^h e^{zw}/Q_n(z)$ is zero whenever $h + r \leq \deg Q_n - 2$, by the residue at infinity. The first possible order is therefore

$$\deg Q_n - h - 1 = N_n + M - h - 1 \geq N_n.$$

Multiplication by x^n does not change this order. Also $\log x = (x-1) + O((x-1)^2)$. □

4.4 A nonzero determinant, including its leading term

Proposition 4.3 (Exact determinant)**.** *There exists $c_n \neq 0$ such that*

$$\det(A_{h,j,n}(x))_{0\le h,j\le m} = c_n x^{\sum_j (n-\beta_j)}(x-1)^{N_n}. \tag{4.7}$$

In particular $A_n(\mathrm{i})$ is invertible.

Proof. The supports give the power of x. After it is removed, the determinant has degree at most N_n. Replace column zero by the sum of column j multiplied by $(\log x)^j$, for $0 \le j \le m$. Its determinant is unchanged; the new column consists of the $R_{h,n}(x)$. Lemma 4.2 supplies the factor $(x-1)^{N_n}$.

To prove nonvanishing, take a block of repeated left endpoints

$$\alpha_u = \cdots = \alpha_v = \alpha, \qquad r = v - u + 1.$$

The pole at $-\alpha$ has multiplicity $v+1$. Write

$$\frac{1}{Q_n(-\alpha+w)} = w^{-(v+1)} B_\alpha(w), \qquad B_\alpha(0) \neq 0.$$

For $u \le j \le v$, the coefficient of $x^{n-\alpha}$ in column j is

$$\frac{C_n}{j!}[w^{v-j}]((-\alpha+w)^h B_\alpha(w)).$$

The orders $v-j$ are $0,\dots,r-1$, in reverse order. Multiplication by B_α gives an invertible triangular change on these functionals.

In all blocks together, the leading-coefficient matrix reduces to the divided-derivative evaluations of $1, z, \dots, z^m$ at distinct points $-\alpha$. This confluent Vandermonde matrix is invertible: a polynomial in its kernel would have zeros of total multiplicity $M = m+1$, exceeding its degree. Thus the leading determinant is nonzero, proving the identity. □

High vanishing order and a degree bound alone would also allow the zero determinant. The leading-term argument rules out that possibility.

5 Prime divisors and weighted zero arcs

5.1 A local valuation inequality

For a rational number $u \neq 0$, let $\mathrm{ord}_p(u)$ be its usual p-adic valuation, and put $\mathrm{ord}_p(0) = +\infty$. We only use primes $\sqrt{n} < p \le n$, so a nonzero distance of absolute value at most n has valuation at most one.

Lemma 5.1 (A coefficientwise divisor)**.** *For such a prime and an integer $0 \le \ell \le n$, put*

$$w_a(p,\ell) = \left\lfloor \frac{\ell_a}{p} \right\rfloor - \left\lfloor \frac{\ell - \alpha_a}{p} \right\rfloor - \left\lfloor \frac{\beta_a - \ell}{p} \right\rfloor.$$

Each w_a is 0 or 1. Every coefficient of $x^{n-\ell}$ in $A_{h,j,n}$ has valuation at least

$$\sum_{a=0}^{m} w_a(p,\ell) + j. \tag{5.1}$$

Proof. The assertion $w_a \in \{0,1\}$ follows from $\lfloor u+v\rfloor - \lfloor u\rfloor - \lfloor v\rfloor \in \{0,1\}$, with $u = (\ell-\alpha_a)/p$ and $v = (\beta_a - \ell)/p$.

For a pole factor, the constant in (4.5) is a binomial coefficient of valuation exactly w_a, by the factorial valuation formula and $p^2 > n$. Its coefficient of order u has valuation at least $w_a - u$.

For a regular factor the constant term has valuation $w_a - 1$. For example, if $\ell < \alpha_a$ and $s = \alpha_a - \ell$, it equals

$$H_a(-\ell) = \frac{\ell_a!(s-1)!}{(s+\ell_a)!}.$$

Use $\lfloor (s-1)/p\rfloor = -\lfloor -s/p\rfloor - 1$; the other side of the interval is identical up to sign. Expansion of the reciprocal linear factors loses at most one valuation per order, so order u has valuation at least $w_a - 1 - u$.

The needed total order is at most $\mu_\ell - 1 - j$, and there are $M - \mu_\ell$ regular factors. Since $\mathrm{ord}_p(d_n) = 1$, the lower bound after multiplication by d_n^m is

$$m + \sum_a w_a - (M - \mu_\ell) - (\mu_\ell - 1 - j) = \sum_a w_a + j.$$

The factor $m!/j!$ and coefficients of $(-\ell + w)^h$ are integers and cannot lower this bound. Sums of terms preserve the lower bound. □

This estimate is stronger than needed. We discard the extra j and minimize over the pole location in order to obtain a divisor common to all entries of the matrix.

5.2 Replacing pole locations by a circle

Write $n = LN_0$. For the grouped intervals define

$$W_j(z,y) = \lfloor (L-2t_j)z\rfloor - \lfloor y - t_j z\rfloor - \lfloor (L-t_j)z - y\rfloor. \tag{5.2}$$

It is periodic in z and y with period one and takes values in $\{0,1\}$. At $z = N_0/p,\ y = \ell/p$, it is precisely the local w_a for an interval of type j.

For generic z, the set on which $W_j(z,y) = 0$ is the closed counterclockwise arc from $\{t_j z\}$ to $\{(L-t_j)z\}$. Its length is $\{(L-2t_j)z\}$, where braces denote fractional part. Thus

$$f(z) = \min_{0\le y<1} \sum_j \nu_j W_j(z,y) = M - \max_y \{\text{total weight of zero arcs containing } y\}. \tag{5.3}$$

The weights are the interval multiplicities ν_j.

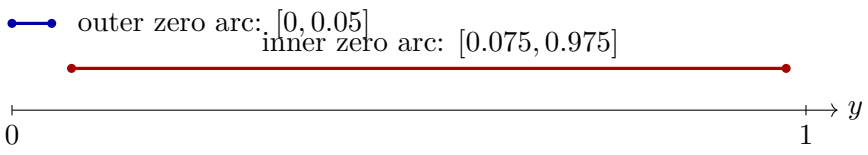


Figure 2: For $L = 14,\ t_0 = 0,\ t_1 = 1,\ z = 3/40$, the zero arcs are disjoint. With equal weights r, their common-complement cost is $f(z) = r$: at least one type contributes r at every y.

All endpoints are affine integer multiples of z, modulo one. Their cyclic order changes only at the following finite rational set. Let

$$\mathcal{S} = \{0\} \cup \{t_j, L - t_j : 0 \le j < J\}, \qquad \mathcal{D} = \{|u - v| : u, v \in \mathcal{S},\ u \ne v\}.$$

Take all fractions $a/d \in [0,1]$ with $d \in \mathcal{D}$, remove duplicates, and list them as

$$0 = u_0 < u_1 < \cdots < u_R = 1.$$

On each open interval (u_r, u_{r+1}), the integer $f(z)$ is constant. At a fixed rational midpoint z, it can be calculated exactly by checking midpoints between consecutive arc endpoints. Off this exceptional set no distinct endpoints coincide, so an endpoint cannot have a smaller value than both adjacent open arcs.

To avoid endpoint conventions entirely, define $f = 0$ at every breakpoint and extend it periodically. This only weakens the local divisor estimate and does not change any integral below.

5.3 The common divisor and its exact exponential rate

Define

$$\Pi_n = \prod_{\sqrt{n}<p\le n} p^{f(N_0/p)}. \tag{5.4}$$

By Lemma 5.1, Π_n divides every integer coefficient of every $A_{h,j,n}(x)$. It is a rational integer; we do not need factorization into Gaussian primes.

Proposition 5.2 (Digamma formula)**.** *The limit* $\gamma = \lim n^{-1} \log \Pi_n$ *exists and is*

$$\boxed{\gamma = \frac{1}{L} \sum_{\substack{0\le r<R \\ u_r>0}} f\left(\frac{u_r + u_{r+1}}{2}\right) (\psi(u_{r+1}) - \psi(u_r)),} \tag{5.5}$$

where $\psi = \Gamma'/\Gamma$.

Proof. For all sufficiently small $z > 0$, none of the endpoints wraps around the circle and the zero arcs are nested about $Lz/2$. They have a common point, so $f(z) = 0$. Thus the apparent singularity at $u_0 = 0$ contributes nothing.

For an interval $(a, b) \subset (0, 1)$ on which $f = v$, periodicity selects primes satisfying

$$N_0/(r + b) < p < N_0/(r + a), \qquad r = 0, 1, \ldots.$$

For each fixed r, the prime number theorem $\vartheta(X) = \sum_{p\le X} \log p = X + o(X)$ makes their contribution to $n^{-1} \log \Pi_n$ tend to

$$\frac{v}{L}\left(\frac{1}{r + a} - \frac{1}{r + b}\right).$$

The sum over $r \ge T$ is bounded, uniformly in large n, by a constant times $M/(LT)$, using the elementary Chebyshev bound $\vartheta(X) = O(X)$. Primes below $\sqrt{n}$ contribute $o(n)$. These observations justify passage from finitely many periods to all periods.

Breakpoint exclusions also cost only $O(\log n)$. Indeed, $\{N_0/p\} = a/d$ implies that p divides dN_0; the denominators d belong to a fixed finite set. Finally,

$$\sum_{r=0}^{\infty}\left(\frac{1}{r + a} - \frac{1}{r + b}\right) = \psi(b) - \psi(a).$$

Summing over the finite partition proves the formula. □

The formula is a lower bound on the content available for division, not a formula for the entire actual gcd of a row after evaluation at $x = \mathrm{i}$. Additional cancellations at i could give more divisibility.

5.4 A transparent two-layer special case

For $t_0 = 0, t_1 = 1$ with weights $r, M - r$, put

$$a_\ell = \max\left(\frac{\ell}{L}, \frac{\ell - 1}{L - 2}\right), \qquad b_\ell = \frac{\ell}{L - 1}, \quad 1 \le \ell \le L - 2.$$

On these intervals the two zero arcs are disjoint, and $f = \min(r, M - r)$. They are precisely the disjointness intervals away from endpoints. Hence

$$\gamma = \frac{\min(r, M - r)}{L} \sum_{\ell=1}^{L-2} (\psi(b_\ell) - \psi(a_\ell)). \tag{5.6}$$

This is a useful small test of the general partition algorithm. For many layers, the weighted maximum-overlap calculation is more effective than checking pairwise disjointness alone.

6 Uniform analytic estimates

6.1 A coefficient bound valid on a whole disk

Lemma 6.1 (Height and divided difference)**.** *Uniformly for $0 \le h, j \le m$,*

$$|A_{h,j,n}(\mathrm{i})| \le \exp(G_0 n + o(n)).$$

After division by Π_n, put

$$U_{h,n}(Y) = \sum_{j=0}^{m} \frac{A_{h,j,n}(\mathrm{i})}{\Pi_n} Y^j \in \mathbb{Z}[\mathrm{i}][Y],$$

and define the polynomial

$$T_{h,n}(Y) = \sum_{j=1}^{m} \frac{A_{h,j,n}(\mathrm{i})}{\Pi_n} \sum_{a=0}^{j-1} \lambda^{j-1-a} Y^a.$$

For $|Y| \le 2$, uniformly in h, Y,

$$|U_{h,n}(Y)|, \ |T_{h,n}(Y)| \le \exp(Gn + o(n)), \tag{6.1}$$

and

$$U_{h,n}(\lambda) - U_{h,n}(Y) = (\lambda - Y) T_{h,n}(Y). \tag{6.2}$$

Proof. In (4.5), the constant has modulus at most 2^{ℓ_a}. For a regular factor the constant has modulus at most one, since it is, up to sign, $\ell_a!(s-1)!/(s+\ell_a)!$ for an integer $s \ge 1$. For either type, the remaining reciprocal linear factors have nonzero integral distances. A coefficient of fixed order at most m is bounded by the corresponding coefficient of $(1 - w)^{-(N_n + M)}$, and is therefore polynomial in n. The factors $(-\ell + w)^h$, and the sum over ℓ, introduce only more fixed powers of n. Thus

$$|A_{h,j,n}(\mathrm{i})| \le K\, d_n^m n^B 2^{\sum_a \ell_a}.$$

Since $\log d_n = n + o(n)$ and $\sum_a \ell_a = n \sum_j \nu_j c_j$, this has rate G_0. Division by Π_n subtracts γ. The degree m and the bounds $|Y| \le 2$, $|\lambda| < 2$ are fixed, so evaluation and the displayed finite difference introduce only constant factors. The identity follows from $\lambda^j - Y^j = (\lambda - Y) \sum_{a=0}^{j-1} \lambda^{j-1-a} Y^a$. □

This is a uniform estimate on a disk, not just an asymptotic value at a single point. Uniformity will permit a rational approximation to select its own nearby algebraic point $Y = y$.

6.2 A global horizontal-contour bound

The main integral estimate can be proved without leaving its tails implicit. First observe that the solution of (1.2) is unique: its left side decreases continuously from $M\pi$ to zero. Also

$$Y > \frac{1}{2}. \tag{6.3}$$

Indeed the outer type has $c_0 = 1, \nu_0 \geq 1$, and $M \geq 2$; at $Y = 1/2$ that type alone contributes at least $\pi/2$, and the remaining multiplicity makes the inequality strict.

Lemma 6.2 (A uniform tail inequality)**.** *For $Y > 1/2$, put*

$$B_Y = \frac{(Y^2 + 1/4)^2}{Y^2 - 1/4}.$$

If $d = \deg Q_n = N_n + M$, then for every real t,

$$\frac{|Q_n(-n/2 + nt + \mathrm{i}nY)|}{|Q_n(-n/2 + \mathrm{i}nY)|} \geq (1 + t^2/B_Y)^{d/2}. \tag{6.4}$$

The same inequality holds on the lower horizontal line.

Proof. The offsets $u = r/n - 1/2$ in each interval occur in pairs $u, -u$, with $|u| \leq 1/2$. For a pair, the square of the modulus ratio is

$$1 + \frac{2(Y^2 - u^2)t^2 + t^4}{(Y^2 + u^2)^2}.$$

It is at least $(1 + t^2/B_Y)^2$: the coefficient of t^2 is at least $2/B_Y$, and that of t^4 is at least $1/B_Y^2$. An unpaired central factor has ratio $\sqrt{1 + t^2/Y^2} \geq \sqrt{1 + t^2/B_Y}$. Multiplication proves the claim, with repetitions included. □

Open the contour in (4.2) to the lines $\operatorname{Im} z = \pm nY$. For each fixed n the vertical ends vanish as their real parts tend to infinity: the rational factor decays faster than $1/|z|$. On these horizontal lines, $|\mathrm{i}^{z+n}| \leq e^{\Theta nY}$. Lemma 6.2 therefore gives

$$|R_{h,n}(\mathrm{i})| \leq K_{h,Y} n^{h+1} e^{\Theta nY} \frac{C_n}{|Q_n(-n/2 + \mathrm{i}nY)|}. \tag{6.5}$$

To justify a constant independent of n, the remaining integral is bounded by

$$\int_{\mathbb{R}} \frac{(|t| + Y + 1/2)^h}{(1 + t^2/B_Y)^{d/2}}\, \mathrm{d}t.$$

For sufficiently large n, replace the denominator exponent by the fixed value $h+2$; the resulting integral is finite. This controls the entire line, including both infinite tails.

6.3 Evaluating the central exponential rate

For one interval of length cn, Stirling's formula and a Riemann sum on a smooth integrand give

$$\log((cn)!) = cn \log n + cn \log c - cn + O(\log n),$$

$$\sum_{r=\alpha}^{\beta} \log|r - n/2 + \mathrm{i}nY| = (cn + 1)\log n + n \int_{-c/2}^{c/2} \log|u + \mathrm{i}Y|\, \mathrm{d}u + O(1).$$

The extra endpoint in the product changes only $O(\log n)$. Direct integration shows

$$\int_{-c/2}^{c/2} \log|u+\mathrm{i}Y|\,\mathrm{d}u = \frac{c}{2}\log(Y^2+c^2/4) - c + 2Y\arctan\frac{c}{2Y}.$$

Subtracting the product logarithm from the factorial logarithm thus gives $-nF_c(Y)+O(\log n)$. Equation (6.5) yields

$$|R_{h,n}(\mathrm{i})| \le \exp\left\{-n\left(\sum_j \nu_j F_{c_j}(Y) - \Theta Y - m\right) + o(n)\right\}.$$

Proposition 6.3 (Small remainders)**.** *After division by Π_n, uniformly in $0 \le h \le m$,*

$$|U_{h,n}(\lambda)| \le \exp(-Dn + o(n)). \tag{6.6}$$

Finally,

$$\frac{\mathrm{d}}{\mathrm{d}Y}F_c(Y) = 2\arctan\frac{c}{2Y}, \qquad \frac{\mathrm{d}^2}{\mathrm{d}Y^2}F_c(Y) = -\frac{c}{Y^2+c^2/4}.$$

Hence (1.2) is the unique global maximizer of the central decay bound. This optimization needs only a one-dimensional monotone root, not a claimed exact asymptotic formula for an oscillating complex sum.

7 From logarithmic polynomials to integer powers

7.1 Why a resultant is the right arithmetic object

The Gaussian integers $\mathbb{Z}[\mathrm{i}]$ form a unique factorization domain. Every nonzero Gaussian integer has complex modulus at least one. If f is a primitive polynomial over this ring and divides a polynomial over $\mathbb{Q}(\mathrm{i})$, Gauss's lemma makes that divisibility valid over $\mathbb{Z}[\mathrm{i}]$ as well.

For polynomials f, U, the resultant vanishes exactly when they have a common root. If f has leading coefficient a and roots $y_1, \dots, y_d$, then

$$\operatorname{Res}(f,U) = a^{\deg U}\prod_{\ell=1}^{d} U(y_\ell).$$

Using a *minimal* polynomial, rather than the whole binomial that first defines y, guarantees the nonzero product needed below.

Lemma 7.1 (Uniform algebraic-value bound)**.** *Let $k \ge 2$, let $p/q > 0$ be reduced, and suppose $|p/q - \pi^k| < 1$. Put*

$$\theta = (p/q)^{1/k} > 0, \qquad y = \mathrm{i}\theta/2.$$

For every n, some h has $U_{h,n}(y) \ne 0$. For each $\eta > 0$ and all sufficiently large n, this h satisfies

$$|U_{h,n}(y)| \ge (2^k q)^{-m}\exp(-(k-1)(G+\eta)n). \tag{7.1}$$

The threshold is independent of p, q.

Proof. Dividing the matrix by Π_n preserves its invertibility. It cannot send the nonzero vector $(1, y, \dots, y^m)^{\mathsf{T}}$ to zero, so some row U is nonzero at y.

Let $f \in \mathbb{Z}[\mathrm{i}][Y]$ be a primitive minimal polynomial of y over $K = \mathbb{Q}(\mathrm{i})$, of degree $d \le k$, and let $a \neq 0$ be its leading coefficient. It divides

$$(2^k q)Y^k - \mathrm{i}^k p$$

over $\mathbb{Z}[\mathrm{i}]$. Thus $a \mid 2^k q$, in particular $|a| \le 2^k q$. Every root of f belongs to $\{y\zeta : \zeta^k = 1\}$. The error assumption implies $|y| < 2$, so all these roots lie in the disk used in Lemma 6.1.

Irreducibility of f and $U(y) \neq 0$ give $\mathrm{Res}(f, U) \neq 0$. Its Gaussian-integrality yields

$$1 \le |a|^{\deg U} |U(y)| \prod_{\ell=2}^{d} |U(y_\ell)|.$$

Use $\deg U \le m$, $d \le k$, and the uniform height bound. Since $G + \eta > 0$, replacing $d - 1$ by $k - 1$ only weakens the result and gives (7.1). □

No assumption of irreducibility of $Y^k - \mathrm{i}^k p/(2^k q)$ has been made. If p/q is a rational power, or a binomial splits over K, the true degree d is simply smaller. In particular, the argument covers every denominator and not just a generic subsequence of approximations.

7.2 The quantitative comparison

Proof of Theorem 1.1. Write $\Delta_k = D - (k-1)G > 0$. Choose $\eta > 0$ small enough that $\Delta_k - k\eta > 0$. All estimates can be made simultaneous: for $n \ge n_0$, every row and every $|Y| \le 2$ satisfy

$$|U_{h,n}(\lambda)| \le e^{-(D-\eta)n}, \quad |U_{h,n}(Y)|,\ |T_{h,n}(Y)| \le e^{(G+\eta)n}.$$

Here n was required to be a multiple of L. Accordingly choose

$$n = L \left\lceil \frac{m \log(2^k q) + \log 2}{L(\Delta_k - k\eta)} \right\rceil. \tag{7.2}$$

For sufficiently large q, this is at least n_0. The extra rounding is at most L, a fixed constant.

Select the row in Lemma 7.1. The ratio of the small remainder to its nonzero value at y is at most

$$(2^k q)^m e^{-(\Delta_k - k\eta)n} \le \tfrac{1}{2}.$$

The divided-difference identity consequently gives

$$|\lambda - y| \ge \tfrac{1}{2}(2^k q)^{-m} e^{-k(G+\eta)n}.$$

If the divided difference were zero, that same identity would contradict the preceding factor $1/2$; hence no division by zero is hidden here.

Substitution of (7.2) yields

$$|\lambda - y| \ge c_{k,m,\eta} q^{-m(D+G)/(\Delta_k - k\eta)}.$$

Indeed $\Delta_k - k\eta + k(G + \eta) = D + G$, which explains the numerator in the final exponent.

Finally,

$$\left|\pi^k - \frac{p}{q}\right| = |\pi - \theta| \sum_{a=0}^{k-1} \pi^{k-1-a}\theta^a \ge 2\pi^{k-1}|\lambda - y|.$$

Fractions outside $|p/q - \pi^k| < 1$, or with $p/q \le 0$, need no special treatment. Letting η tend to zero proves the irrationality-exponent assertion. □

7.3 Why a common divisor helps twice

Before division, the rates are D_0, G_0. Dividing all coefficients and remainders by the same Π_n changes them to

$$D = D_0 + \gamma, \qquad G = G_0 - \gamma.$$

Thus the numerator $D + G$ is unchanged, while the useful gap grows by $k\gamma$:

$$D - (k-1)G = D_0 - (k-1)G_0 + k\gamma.$$

This is why a divisor must be applied consistently to the remainder, the coefficients, and the divided difference. Reducing only one height in the final formula would give an incorrect improvement.

8 Concrete parameters and every fixed integer power

8.1 Recovering the balanced benchmark

If all M intervals equal $[0, n]$, the arc minimum used here is zero, so $\gamma = 0$ for this particular divisor. This does not prove that the entire actual gcd has rate zero. The root equation becomes

$$Y = \tfrac{1}{2} \cot \frac{\pi}{4M}.$$

Substitution gives

$$D_0 = -M \log \left(2 \sin \frac{\pi}{4M} \right) - m, \qquad G_0 = m + M \log 2.$$

With $m = 289$, the resulting exponent is $2028.145382310168\ldots$. This is a useful check of the normalization and contour calculation.

8.2 Two, seven, and twelve layers

Three configurations illustrate the improvement:

Configuration	m	L	Offsets and multiplicities
Two layers	255	14	$t = (0, 1),\ \nu = (128, 128)$
Seven layers	199	39	$t = (0, \ldots, 6),\ \nu = (80, 20, \ldots, 20)$
Twelve layers	186	69	$t = (0, \ldots, 11),\ \nu = (66, 11, \ldots, 11)$

Their rates for $k = 3$, rounded here only for readability, are:

Configuration	D_0	G_0	γ	E_3
Two layers	955.116623	419.770987	27.777682	1762.608232
Seven layers	664.537005	322.700112	44.701426	1282.033606
Twelve layers	605.861295	301.032295	45.436001	1203.972439

The double-layer result follows also from (5.6). For the twelve-layer case the corrected rates are

$$\begin{aligned} D &= 651.297296270676769392603641659 9\ldots, \\ G &= 255.596293834458504937833052789 8\ldots, \\ D - 2G &= 140.104708601759759516937536080 1\ldots. \end{aligned}$$

The strict conclusion is $E_3 < 1203.973$, not a conclusion inferred from the rounded entries of the table.

These examples arose from bounded searches and a limited multiplicity refinement. They are not the solution of a globally optimized variational problem. In particular, more layers are not known to be necessary or optimal.

8.3 The accepted L99 instance, step by step

Use $L = 99$, offsets $t_j = j$, and base multiplicities

$$v_0 = 16, \qquad v_j = 1 \ (1 \le j \le 27), \qquad c_j = \frac{99 - 2j}{99}.$$

The total base multiplicity and length are

$$V = 16 + 27 = 43, \qquad S = \sum_{j=0}^{27} v_j c_j = 16 + 27 - \frac{2}{99}\frac{27 \cdot 28}{2} = \frac{389}{11}.$$

At scale $s = 4$, this gives $M = 172$, $m = 171$, and total relative interval length $4S = 1556/11$. There are twenty-eight distinct interval types, rather than twenty-eight rows: repeated types account for the much larger matrix dimension 172.

The formal proof uses the fixed rational height

$$Y = \frac{1801}{20} = 90.05.$$

This is close to the maximizing height, but need not be the exact root of (1.2). The contour proof is valid at any fixed $Y > 1/2$; a lower bound on the decay at a convenient height is enough. The one-sided certified estimates are

$$\begin{aligned} D_0(Y) &\ge \frac{9789173}{20000} = 489.45865, \\ G_0 &\le \frac{13452441}{50000} = 269.04882, \\ \gamma &\ge \frac{2761813}{50000} = 55.23626. \end{aligned}$$

Here the last line denotes an available divisor rate: for every positive slack the actual divisor eventually has at least this exponential rate minus the slack. Equality with the exact digamma sum is not needed.

Adding the lower divisor rate to the decay bound and subtracting it from the growth bound gives the usable rational pair

$$D = \frac{54469491}{100000}, \qquad G = \frac{2672657}{12500}.$$

These are deliberately conservative rates, not decimal truncations asserted to equal the exact analytic optimum. Their final use can be checked entirely with integer arithmetic:

$$\begin{aligned} D + G &= \frac{75850747}{100000}, \\ D - 2G &= \frac{54469491 - 2 \cdot 21381256}{100000} = \frac{11706979}{100000}, \\ 171\frac{D+G}{D-2G} &= \frac{171 \cdot 75850747}{11706979} = \frac{12970477737}{11706979}. \end{aligned}$$

For the stated upward rounding, cross multiplication gives

$$110793 \cdot 11706979 - 100 \cdot 12970477737 = 3550647 > 0.$$

This proves the strict comparison with $110793/100 = 1107.93$ without relying on a printed decimal expansion.

Where the divisor number comes from. For the base weights the accepted finite-window calculation gives $\gamma_0 = 2761813/200000 = 13.809065$. Its exact endpoint table has 3004 open cells, of which 3000 carry a positive certified lower cost. The prime-period prefix covers $r = 0, \ldots, 31$; the finite tail covers $r = 32, \ldots, 999999$. Their lower bounds are

$$\text{prefix} \geq \frac{1380525198339888}{10^{14}}, \qquad \text{tail} \geq \frac{381339545481}{10^{14}}.$$

The sum is 13.80906537885369, leaving positive room above 13.809065 for the interval-shrinking loss. Nonnegative periods outside this finite window can be discarded. Scaling all weights by four scales the certified divisor rate by four as well. Section 10 explains how the finite inequalities are connected to the actual asymptotic divisor.

Why the Arb and Lean decimals differ slightly. The earlier Arb calculation at the optimizing height gives an expression near 1107.9267863. The accepted exact rational expression is $1107.9269670681\ldots$. The small increase comes from a rational trial height, finite divisor windows, and one-sided analytic estimates. The theorem claims the latter number. A sharper experimental decimal is not substituted into an accepted theorem.

8.4 An explicit scale recipe for every fixed power

For the twelve-layer shape let

$$v_0 = 6, \quad v_j = 1 \ (1 \leq j \leq 11), \quad c_j = 1 - \frac{2j}{69}, \quad \nu_j = s v_j.$$

Then

$$M = 17s, \quad m = 17s - 1, \quad S = \sum_j v_j c_j = \frac{347}{23}.$$

The arc partition is unchanged when all weights are multiplied by s, so

$$\gamma = s\gamma_*, \qquad \gamma_* = 4.130545573141366136266461634 1\ldots.$$

Introduce the fixed constants

$$g_* = 17 + S\log 2 - \gamma_*, \qquad g_* = 23.326935803132591357984822980 8\ldots, \tag{8.1}$$

$$A_* = \sum_j v_j c_j \log c_j,$$

$$d_* = S\log(S/\Theta) - A_* - 17 + \gamma_*. \tag{8.2}$$

Since $f \leq 17$ for the base weights, the prime number theorem gives $\gamma_* \leq 17$. Thus $g_* \geq S\log 2 > 1$, proving $G = sg_* - 1 > 0$ without requiring a decimal estimate. The certificate more precisely verifies $g_* > 23$.

Proposition 8.1 (A root-free admissibility recipe)**.** *For any fixed integer $k \geq 2$, let*

$$s_k = \max\left\{1,\ \left\lceil \exp\left(\frac{(k-1)g_* - d_*}{S} + 1\right)\right\rceil\right\}. \tag{8.3}$$

This choice satisfies $D - (k-1)G > 0$. More explicitly, put

$$\underline{D}_s = s(S\log s + d_*) + 1 - \frac{\Theta^2}{12sS}, \qquad G_s = sg_* - 1.$$

Whenever $\underline{D}_s > (k-1)G_s$, the following bound needs no numerical root Y:

$$\mu(\pi^k) \leq (17s-1)\frac{\underline{D}_s + G_s}{\underline{D}_s - (k-1)G_s}. \tag{8.4}$$

In particular this is valid at $s = s_k$.

Proof. Evaluate the decay expression at the trial height $Y_0 = sS/\Theta$; the optimized D is at least this trial value. For every $x \geq 0$, $\arctan x \geq x - x^3/3$ and $\log(1+x^2) \geq 0$. Consequently

$$F_c(Y_0) \geq c\log(Y_0/c) + c - \frac{c^3}{12Y_0^2}.$$

Since $0 < c_j \leq 1$, one has $\sum_j v_j c_j^3 \leq S$. The terms $sS - \Theta Y_0$ cancel, giving

$$D \geq s(S\log s + d_*) + 1 - \frac{\Theta^2}{12sS} = \underline{D}_s.$$

At $s = s_k$,

$$\underline{D}_s - (k-1)G_s \geq sS + k - \frac{\Theta^2}{12sS} > 0.$$

Here $S = 347/23$, $s \geq 1$, and $\Theta < 2$, so the last positivity is immediate.

For $G_s \geq 0$ the function $m(D+G_s)/(D-(k-1)G_s)$ decreases as D increases. Replacing D by the lower bound $\underline{D}_s$ therefore gives (8.4). □

This proposition supplies a definite upper-bound recipe for *every fixed integer power $k \geq 2$*, in addition to the optimized examples in Corollary 1.3. It does not require an unbounded parameter search to establish admissibility.

8.5 Why the higher-power constants still grow rapidly

Expanding the root equation for large s gives

$$Y = \frac{sS}{\Theta} + O(s^{-1}), \quad D = s(S\log s + d_*) + 1 + O(s^{-1}), \quad G = sg_* - 1.$$

Thus the threshold for s grows exponentially with k. The explicit choice (8.3) yields

$$E_k = O(k\rho^k), \qquad \rho = \exp(g_*/S) = 4.69343992780744986545\ldots. \tag{8.5}$$

To justify the order estimate directly, that choice has $s_k = O(\rho^k)$, a gap at least $s_k S + O(1)$, and numerator $(17s_k - 1)(\underline{D}_{s_k} + G_{s_k}) = O(ks_k^2)$. All implied constants are fixed by this twelve-layer shape. Equation (8.5) describes the construction as k varies; it must not be confused with the separate limit $n \to \infty$ used to prove each fixed-k bound.

For orientation, the certified optimized values before upward rounding are

k	E_k
3	1203.97243920634698117...
5	42460.70466848845471678...
7	1272102.82718073869325320...
9	35398936.33162709355243184...

The number s was optimized only within one fixed shape.

8.6 A simple all-power bound from the L99 shape

The same root-free calculation can use the rational lower divisor rate already accepted for the L99 base shape. Put

$$V = 43, \quad S = \frac{389}{11}, \quad \gamma_0 = \frac{2761813}{200000}, \quad A = \sum_{j=0}^{27} v_j c_j \log c_j,$$

and define

$$g = V + S\log 2 - \gamma_0, \qquad d = S\log(S/\Theta) - A - V + \gamma_0.$$

For every positive integer scale s, the available growth rate and trial-height decay lower bound are

$$G_s = sg - 1, \qquad \underline{D}_s = s(S\log s + d) + 1 - \frac{\Theta^2}{12sS}. \tag{8.6}$$

To obtain these formulas, repeat the preceding proof with $V = 43$ in place of 17, and use the lower divisor rate $s\gamma_0$ consistently on both sides. No new arc partition is needed when only the common scale changes.

Proposition 8.2 (A coarse uniform paper-level bound)**.** *For every integer $k \geq 3$, the L99 shape with $s = 8^k$ gives*

$$\mu(\pi^k) < 301 \cdot 8^k.$$

This is a consequence of the written estimates and the available scaled-family interface; it is not an accepted all-k Lean theorem.

Proof. The elementary inequalities $2/3 < \log 2 < 3/4$, $35 < S < 36$, and $13 < \gamma_0 < 43$ imply

$$1 < g < 43 + 36(3/4) - 13 < 60.$$

Since $c_j \leq 1$, one has $A \leq 0$. Also $S/\Theta > e$, using $\Theta < 2$ and $e < 3$, and $S - 43 + \gamma_0 > 0$. Hence $d > 0$.

For $s = 8^k$, $\log s = 3k\log 2$, so $S\log s > 70k$. Write $e_s = \Theta^2/(12sS)$, with $0 < e_s < 1$. Equation (8.6) gives

$$\begin{aligned}\Delta_s &= \underline{D}_s - (k-1)G_s \\ &= s\{S\log s + d - (k-1)g\} + k - e_s > 10ks.\end{aligned}$$

Moreover $0 < G_s < 60s$, and $m = 43s - 1 < 43s$. Therefore

$$m\frac{\underline{D}_s + G_s}{\Delta_s} = m\left(1 + \frac{kG_s}{\Delta_s}\right) < 43s(1+6) = 301 \cdot 8^k.$$

The trial height sS/Θ is greater than $1/2$, so the contour estimate applies. The transfer theorem completes the proof. □

This simple formula avoids an open-ended search to establish the all-power quantifier. It is intentionally loose: even at $k = 3$ it is much larger than the accepted optimized value. Its exponential growth is also worse than the classical $B(k) = O(k \log k)$ in Proposition 3.1. Its value is a transparent route from one certified interval shape to a uniform statement, rather than a new asymptotic record.

8.7 Sensitivity of the bound and safe numerical substitution

Fix m, k and write $\Delta = D - (k-1)G > 0$. Direct differentiation gives

$$\frac{\partial E_k}{\partial D} = -\frac{mkG}{\Delta^2}, \qquad \frac{\partial E_k}{\partial G} = \frac{mkD}{\Delta^2}.$$

When $G \geq 0$, a lower bound for D and an upper bound for G therefore give a safe upper bound for E_k, provided the conservative gap stays positive. Close to a zero gap the same formulas show why a small rate error can produce a large exponent error. Displaying many decimal places does not repair an unjustified estimate on the wrong side.

If only the available divisor rate changes, with D_0, G_0 fixed, then

$$\frac{\mathrm{d}E_k}{\mathrm{d}\gamma} = -\frac{km(D_0 + G_0)}{(D_0 - (k-1)G_0 + k\gamma)^2}.$$

This makes the arithmetic gain quantitative. By contrast, adding layers also changes m, D_0, G_0, so its net effect has to be recomputed. More divisibility alone is not an optimization criterion.

8.8 What would be needed to reach 10–100?

In the present transfer formula, $G \geq 0$ implies

$$E_k = m\left(1 + \frac{kG}{D - (k-1)G}\right) \geq m.$$

The accepted L99 example has $m = 171$. Hence this particular parameter choice cannot possibly give $E_3 < 100$. This is not an impossibility theorem for all nested shapes.

A more substantial change would keep the *target polynomial degree* fixed at three. If

$$|P(\lambda)| \geq c_\varepsilon \mathrm{H}(P)^{-\nu-\varepsilon} \quad (P \in \mathbb{Z}[\mathrm{i}][Y],\ \deg P \leq 3,\ P \neq 0),$$

then taking $P(Y) = 8qY^3 + \mathrm{i}p$ gives

$$|P(\lambda)| = q|\pi^3 - p/q|, \qquad \mu(\pi^3) \leq \nu + 1.$$

Only this sparse binomial family is actually needed. The thresholds $\nu \leq 99$ and $\nu \leq 9$ correspond to upper bounds 100 and 10. Type II Hermite–Padé and multiple orthogonal polynomial constructions are plausible research tools; examples of bounded-degree logarithmic approximation appear in Marcovecchio [3]. Their numerical constants at other arguments cannot be transplanted to $\log \mathrm{i}$ without checking the complex endpoint and its arithmetic.

9 How the constants are computed and certified

9.1 An exact finite combinatorial calculation

The common divisor does not require numerical integration over a circle. The calculation is finite:

1. Form the integer slopes $0, t_j, L - t_j$ and all nonzero differences.
2. Form and sort the rational critical points $a/d \in [0, 1]$.
3. In each open critical cell choose its rational midpoint z. Sort all fractional endpoints $\{t_j z\}, \{(L - t_j)z\}$.
4. Evaluate the integer vector $W(z, y)$ at each rational midpoint between consecutive endpoints. Minimize its weighted sum.
5. Apply the finite digamma sum (5.5). Adjacent cells with the same minimum may be merged.

Only the final special-function evaluations use real arithmetic. All endpoint orderings and minima use exact fractions and integers.

For the base weights $(6, 1, \ldots, 1)$ at $L = 69$, the first merged nonzero cells are:

Left endpoint	Right endpoint	f
1/69	1/67	6
1/67	1/65	7
1/65	1/62	8
1/62	1/60	7
1/60	1/58	6
1/58	1/56	5
1/56	1/54	4
1/54	1/52	3

These are only the beginning of the partition. Endpoint wraparound and changing overlaps make a full computation preferable to extrapolating a pattern from the first few cells. The complete merged list is supplied in the certificate data.

9.2 Ball arithmetic, root signs, and strict rounding

The standalone program `code/certify.py` requires Python and `python-flint`. It uses Arb [6] at 256-bit precision. A ball encloses the exact real value by a midpoint and a rigorously propagated error radius.

The saddle equation is strictly decreasing. At $Y = 1/2$ its left side minus Θ is positive, and at $Y = 2M$ it is negative. The program performs 210 rational bisections, accepting each sign only if the entire corresponding ball has that sign. It then evaluates $D_0, G_0, \gamma, D, G, \Delta_k, E_k$ on the resulting interval and asserts both $\Delta_k > 0$ and the advertised strict upper bound.

For example, in the twelve-layer cubic case it encloses

$$E_3 = 1203.97243920634698117089090450914656\ldots$$

with an error radius below $6 \cdot 10^{-59}$. The assertion $E_3 < 1203.973$ has a wide safety margin relative to this radius. The certificate does not assume that a floating-point optimizer located a true minimum.

The same program checks the arc bound independently against all integer pole positions for $n = 345$: 52 nonexceptional primes are checked, and 39 have a positive guaranteed divisor exponent. This finite check tests the combinatorial implementation; the proof for arbitrary n is Lemma 5.1.

9.3 Reproduction and the role of each file

Run the following from the project directory:

```
python code/certify.py
python code/finite_checks.py
xelatex -interaction=nonstopmode -halt-on-error main.tex
xelatex -interaction=nonstopmode -halt-on-error main.tex
```

The first command regenerates `data/certificates.json`. The second requires SymPy and checks the small exact examples in Appendix A. Optional experiment programs require the additional packages stated in the README. They are not prerequisites for the main proof.

The certificate records the parameters, merged nonzero arc cells, root enclosure, all rates, the gap, and the strict comparisons. Its purpose is to make the numerical part reproducible. The prime number theorem and the asymptotic inequalities are mathematical inputs, not claims checked by a finite program.

For the revised L99 ledger and fourth-power comparison, additionally run

```
python code/check_current_results.py
```

This short program checks the exact rational exponent and its upward rounding, encloses the analytic expressions at the fixed rational height, and checks the root-free fourth-power expression with Arb. It uses the bundled copy of the executed L99 acceptance report to identify the accepted constants. It neither reruns Lean nor treats that report as a new proof. The finite divisor table and the asymptotic interface are justified by the accepted source closure, not by this numerical replay. Its output is `data/current_results_2026-10-31.json`.

The original `certify.py` is retained as the reproducible certificate for the earlier odd-power table. Its odd-k input restriction reflects the scope of those historical examples, not a parity restriction on Theorem 1.1.

9.4 Parameter order and effectivity

For a chosen integer $k \geq 2$, first fix L, t_j, ν_j, or fix the scale s in the twelve-layer family. Then fix an exponent slack ε, and choose the small η in Section 7. Only after these choices does n tend to infinity, and only then is n selected as a function of the approximation denominator q.

This manuscript gives explicit upper bounds on the irrationality *exponent*. It does not tabulate an explicit numerical $q_0(\varepsilon)$ or the constants in (1.1). Those would require explicit versions of the prime-counting remainder and all finite error terms. They are a separate effectivity calculation, not part of the decimal certification.

10 What has been formalized, and how the proof is organized

10.1 The exact accepted statement

The latest fixed theorem is named

```
OddPiPowers.L99.pi_cube_exponentUpperBound_l99_exact.
```

Its conclusion is

```
ExponentUpperBound (Real.pi ^ 3) (12970477737 / 11706979)
```

The predicate used by the project expands to

$$\begin{aligned} &\forall \varepsilon > 0\ \exists c > 0\ \exists Q \in \mathbb{N}\ \forall p \in \mathbb{Z}\ \forall q \in \mathbb{N}, \\ &\quad q > 0,\ q \geq Q \quad \Longrightarrow \quad cq^{-(E+\varepsilon)} \leq |\xi - p/q|. \end{aligned} \tag{10.1}$$

Every integer numerator and every sufficiently large positive denominator are included; reduced fractions are not required. For an irrational ξ, this implies the usual $\mu(\xi) \leq E$. The project also connects the predicate to Mathlib's infinitely-often approximation predicate `LiouvilleWith` and proves the corresponding strict-bound corollary at 110793/100.

The all-fractions formulation is useful for more than convenience. If $\xi = a/b$ were rational, the same value would be represented by $p = ta$, $q = tb$ for arbitrarily large t. The right side of (10.1) would then be zero while its left side is positive. Thus this interface itself proves irrationality. There is no loophole in which equal fractions are silently omitted.

10.2 From an abstract transfer to the actual family

A transfer theorem with hypotheses about small polynomials is only one part of the task. It must be supplied with the specific polynomials obtained from the integral construction. For the accepted instance the dependency chain contains the following mathematical interfaces:

1. The integral and local coefficient formulas define integer polynomials, with the required support and high-order vanishing.
2. The exact determinant gives an invertible coefficient matrix after evaluation at i, and after division by a nonzero common integer divisor.
3. Coefficientwise prime valuations show that the chosen integer really divides every coefficient. The reduced Gaussian rows are therefore integral objects, rather than formal quotients over a field.
4. Analytic bounds supply uniform height, divided-difference, and remainder estimates for these actual rows.
5. The finite-window prime estimate gives the required asymptotic lower bound for their actual divisor.
6. One-sided rational numerical estimates assemble an `AuxiliaryFamily` with $m = 171, L = 99$ and the displayed rational rates. The accepted general transfer is then applied at $k = 3$.

The distinction between a named hypothetical family and the actual integral family was a substantial part of the formalization work. The final accepted statement closes that connection.

10.3 Why a midpoint table needs a real-variable proof

The exact combinatorial program finds cells by sorting rational crossings of affine endpoint functions. A finite midpoint calculation identifies a candidate order and cost in each cell. By itself this is a test at one point. To turn it into a theorem for all real z in the cell, the formal certificate records floor values, endpoint rank and inverse-rank codes, and the candidate minimum.

The proof checks adjacent endpoint inequalities and permutation consistency. No order change is possible without an endpoint crossing, and all crossing positions are among the cell boundaries. The cost is checked on each gap and at the endpoints of the inner minimization variable y. This last clause prevents an open-gap argument from overlooking a smaller cost exactly at an endpoint. The result is a lower bound valid for every relevant y and every real z in each open cell. The L99 table has 3004 cells and 56 labeled endpoints before multiplicities are expanded.

This explains the difference between the very short search code and the larger proof data. The code proposes a finite answer; the checked certificate supplies the order and cost facts needed to justify that answer on a continuum. The breakpoint convention still assigns zero to the finitely many outer exceptional points, as in Section 5.

10.4 How finite windows prove an infinite asymptotic estimate

The paper formula for γ uses a digamma difference, hence all prime periods. The accepted fixed proof needs only a lower rate, so it uses finitely many periods and drops the rest. For a cell (a, b), an inward shrink by a positive rational fraction avoids endpoint ambiguities. For each retained period r, the prime number theorem gives the contribution of

$$\frac{N_0}{r+b'} < p < \frac{N_0}{r+a'}, \qquad a < a' < b' < b.$$

The limit of its normalized logarithmic prime product is $L^{-1}((r+a')^{-1} - (r+b')^{-1})$, multiplied by the certified cell cost. Summing a fixed finite set of such estimates preserves their eventual validity.

For L99 the relative shrink is $\epsilon = 10^{-56}$. The symbolic loss allowance is

$$\frac{2 \cdot 99^2 \epsilon \cdot 1000000 \cdot 43 \cdot 3004}{99}.$$

The 32 short prefix periods are checked explicitly; the remaining finite range up to one million periods is bounded by a rational telescoping estimate. The compiler does not enumerate a product over all primes or over a million periods. The prime number theorem is used as a proved asymptotic input, and the retained rational sum is enough to exceed the chosen lower rate.

10.5 Why the numerical proof does not trust floating point

The fixed height 1801/20 reduces the analytic comparisons to inequalities involving logarithms and arctangents at explicit rational arguments. A useful logarithm identity is

$$\log x = 2 \sum_{r=0}^{\infty} \frac{1}{2r+1} \left(\frac{x-1}{x+1} \right)^{2r+1}, \qquad x \geq 1.$$

Every term is nonnegative, so truncation gives a rational lower bound. Range reduction by a known power of two keeps the series argument small. The L99 certificates use $4096 = 2^{12}$ for the

central logarithms. For the arctangent contribution, the elementary bound $\arctan u \geq u - u^3/3$, $u \geq 0$, is enough at this large height.

Consequently the analytic lemma reduces to short rational inequalities for the 28 interval types, together with proved monotonicity and series bounds. Ordinary kernel-checked arithmetic then proves the advertised one-sided rates. The earlier Arb calculation remains useful for selecting parameters and checking the size of margins, but is not an assumption of the accepted Lean theorem.

10.6 Acceptance evidence and its limits

The executed acceptance report is

`formalization/OddPiPowers/logs/l99-acceptance-2026-10-31.json.`

It records 147 new dependency modules, four final signature checks, and 15 selected transitive axiom reports, including the actual family and final numerical theorem. The reported logical dependencies are `propext`, `Classical.choice`, and `Quot.sound`. No project mathematical axiom, `sorryAx`, or native-evaluation axiom occurs in the accepted audited closure. These are local executed verification records, not a claim of independent specialist review.

Assertion	Status in this revision
General transfer for integer $k \geq 2$	Accepted Lean theorem
L99 cubic bound $\frac{12970477737}{11706979}$	Accepted fixed Lean application
Earlier P6 cubic bound 42170551935/35026168	Accepted fixed Lean application
Twelve-layer numerical cases $k = 5, 7, 9$	Manuscript argument and Arb certificates
L99 fourth-power comparison at scale 19	Manuscript specialization and Arb comparison
Uniform bound $301 \cdot 8^k$	Paper-level proof; generic Lean assembly remains
Classical consequence $B(k)$	Derived here from a cited external theorem

The earlier P6 exact value is approximately 1203.97275360, slightly above its optimizing Arb value 1203.97243921. The same distinction between conservative formal rates and an optimized analytic expression applies to L99. Both earlier sources and their accepted evidence are retained.

Proof trials were serialized under the project's fixed memory contract. Large table reductions were split into small modules and structural append arguments; the numerical and cell bounds were not weakened to make the checker finish. This organization matters for reproducibility on a machine with limited memory, but it does not change the mathematical statement or the toolchain. This editorial revision does not modify the accepted Lean sources or claim new compiler acceptance for the additional exposition.

A Small examples and useful background

A.1 A complete two-by-two system

Take $m = 1, n = 4$, with intervals $[0, 4]$ and $[1, 3]$. Here $d_4 = 12$, the interval lengths are $4, 2$, and

$$C_n = 1! \cdot 12 \cdot 4! \cdot 2! = 576.$$

The exact coefficient matrix is

$$A_{1,4}(x) = \begin{bmatrix} 4x^4 + 112x^3 - 112x - 4 & -48x^3 - 144x^2 - 48x \\ -160x^3 - 144x^2 + 288x + 16 & 48x^3 + 288x^2 + 144x \end{bmatrix}.$$

Its determinant is

$$\det A_{1,4}(x) = 192x(x-1)^6.$$

The exponent $6 = 4 + 2$ is N_n, while the factor x is $x^{(4-4)+(4-3)}$. The two remainders are

$$R_{0,4}(x) = A_{0,0,4}(x) + A_{0,1,4}(x) \log x, \quad R_{1,4}(x) = A_{1,0,4}(x) + A_{1,1,4}(x) \log x.$$

Their first possible orders at $x = 1$ are respectively seven and six, in agreement with the residue-at-infinity calculation. At $x = \mathrm{i}$ the determinant is nonzero.

This example detects a normalization error immediately. Using $d_4/(4!2!) = 1/4$ instead of 576 would divide the whole matrix by 2304. Its x^4 coefficient in the upper-left entry would be $1/576$, rather than the integer 4.

A.2 Why the minimal polynomial cannot be skipped

Consider $y = \mathrm{i}$ and $U(Y) = Y^2 + \mathrm{i}Y - 1$. Then $U(\mathrm{i}) = -3 \neq 0$, but

$$Y^3 + \mathrm{i} = (Y - \mathrm{i})(Y^2 + \mathrm{i}Y - 1).$$

The product of U over all three roots of $Y^3 + \mathrm{i}$ is zero. Nonvanishing at one root alone would not justify a nonzero full-binomial norm. The minimal polynomial of y over $\mathbb{Q}(\mathrm{i})$ is $Y - \mathrm{i}$; its resultant with U is -3. The same logical issue arises for perfect-power rational approximations. Lemma 7.1 handles it uniformly.

A.3 Divisibility before and after evaluation

A prime dividing every coefficient of a polynomial also divides its value at i in $\mathbb{Z}[\mathrm{i}]$. The converse fails. Evaluation can create cancellations, and an individual p-adic unit coefficient need not remain visible in the sum. For example, $P(x) = 1 + x^2$ has unit coefficients at every odd prime, but $P(\mathrm{i}) = 0$. Less extreme cancellations can raise valuations without making the value zero. Thus boundary coefficient calculations alone cannot bound the gcd of an evaluated row from above.

Our balanced arc minimum $f = 0$ likewise proves only that the particular coefficientwise divisor has $\gamma = 0$. It is not a proof that every possible evaluated-row divisor has rate zero.

A.4 The least common multiple as an arithmetic cost

Prime factorization gives

$$\log d_n = \sum_{p^a \leq n} \log p = n + o(n).$$

The last equality uses the prime number theorem. Thus d_n^m has exponential cost m, although it is much smaller than $(n!)^m$. The interval factorials in C_n, by contrast, cancel the factorial scale of the contour denominator. Their position is crucial before Stirling's formula is applied.

Factors such as $m!$ can be enormous but are fixed as n grows. Their logarithms divided by n vanish. For large m, small-n regressions may confuse these constants with exponential improvement. Exact identities avoid this ambiguity.

A.5 A two-layer arc calculation by hand

For $L = 14, z = 3/40$, the outer zero arc is $[0, \{14z\}] = [0, 1/20]$, and the inner one is $[\{z\}, \{13z\}] = [3/40, 39/40]$. They are disjoint, as drawn in Figure 2. With weights $128, 128$, at least one type contributes 128 at every pole location. Formula (5.6) collects all such intervals and every periodic prime range.

B Finite experiments and further directions

These computations distinguish analytic convergence from arithmetic usefulness. None proves impossibility for its entire method family.

B.1 Scalar Padé approximation to a pure cubic target

The rational-coefficient function

$$F(z) = \frac{\arcsin\sqrt{z}}{\sqrt{z}} = \sum_{j\geq 0} \frac{\binom{2j}{j}}{4^j(2j+1)} z^j$$

satisfies $F(1/4)^3 = \pi^3/27$. An exact diagonal $[n/n]$ Padé approximant to F^3, evaluated at $1/4$ and multiplied by 27, yields a pure π^3 approximation. After complete rational reduction:

n	$\log Q_n/n$	$-\log\lvert Q_n\pi^3 - P_n\rvert/n$
8	37.2732	−32.0818
16	75.4662	−70.2532
32	155.3497	−150.1009

The denominator growth in these samples overwhelms the analytic error reduction, so they do not give shrinking integer linear forms. An approximate quadratic trend in $\log Q_n$ is only an observation.

B.2 A derivative kernel isolating $\beta(3)$

For integers $a \geq 3, n \geq 1$, define

$$\mathcal{R}_n(t) = \frac{(-1)^n(an)!\,4^{3n}}{(n!)^3} \frac{\prod_{j=0}^{n-1}(t-j)^3}{\prod_{k=0}^{an}(t+k+1/2)}.$$

It is proper with simple poles, and

$$\mathcal{R}_n(t) = \sum_{k=0}^{an} \frac{(-1)^k b_k}{t+k+1/2}, \quad b_k = \binom{an}{k}\left(\frac{(2k+2n)!\,k!}{(2k)!\,(k+n)!\,n!}\right)^3.$$

The factorial ratio is integral: it is $4^n(k+1/2)_n/n!$, the coefficient of z^n in

$$(1-4z)^{-k-1/2} = (1-4z)^{-k}(1-4z)^{-1/2},$$

whose two factors have integral coefficients.

Two derivatives leave only cubic poles. Hence

$$\mathcal{S}_n = \frac{1}{2}\sum_{t=0}^{\infty}(-1)^t\mathcal{R}_n''(t) = q_n\beta(3) - p_n,$$
$$q_n = 8\sum_{k=0}^{an} b_k, \qquad p_n = \sum_{k=0}^{an} b_k \sum_{\ell=0}^{k-1} \frac{8(-1)^\ell}{(2\ell+1)^3}.$$

The series is absolutely convergent, since $\mathcal{R}_n''(t) = O(t^{-3})$ even at $a = 3$. The triple zeros make the summands at $0, \dots, n-1$ zero. No $\beta(1)$ or $\beta(2)$ term occurs.

Nevertheless, clearing the actual denominator of p_n and cancelling the complete integer gcd does not make the tested forms small. For $a = 3$:

n	$\log Q_n/n$	$-\log\lvert Q_n\beta(3) - P_n\rvert/n$
8	24.3881	−14.7970
20	25.5856	−15.8243
40	26.4280	−16.7265

The sampled indices up to 40, for $a = 3, \dots, 8$, all have negative decay indicators. This rejects the initial parameter choices as presently tested, not every related kernel.

B.3 What a successful binary family would have to prove

Suppose a new construction supplies nonzero integer linear forms

$$L_n = Q_n\beta(3) - P_n, \quad |L_n| = e^{-\delta n + o(n)}, \quad |Q_n| \le e^{\kappa n + o(n)}, \quad \delta > 0.$$

With nonproportionality allowing a determinant comparison at every denominator scale, one obtains $\mu(\beta(3)) \le 1 + \kappa/\delta$. Targets 100 and 10 correspond to $\delta/\kappa \ge 1/99$ and $\delta/\kappa \ge 1/9$. The derivative experiment isolates the target correctly but has not reached $\delta > 0$.

A next kernel should preserve isolation while improving factorial-ratio divisibility, possibly through related hypergeometric representations. A second route is the fixed-degree polynomial lower bound in Section 8. Both require new arithmetic estimates: a small residual before denominator clearing does not suffice.

B.4 Corrections that should not become unproved claims

A pointwise local-product limit is not a global steepest-descent proof. For example, replacing the balanced G_0 by

$$G_0 + M\log\cos\frac{\pi}{4M}$$

at $m = 289, M = 290$ changes it by only $-0.001063536257139\ldots$. Even if justified, direct substitution gives $2028.126464240039\ldots$, not roughly 2018. No such sharpened oscillatory estimate is used here.

Finite gcd samples also do not prove that an exponential divisor is present or absent. Our main prime-divisor lower bound is proved coefficientwise and does not depend on a gcd regression.

C Notation, intermediate checks, and reading exercises

C.1 A notation sheet

Symbol	Meaning
k	Fixed target exponent in π^k.
L, t_j	Denominator of the interval shape and its distinct offsets.
ν_j	Multiplicity of interval type j.
M, m	Total multiplicity $M = \sum_j \nu_j$; logarithmic degree $m = M - 1$.
$n = LN_0$	Unbounded approximation index; N_0 is a positive integer.
c_j	Relative interval length $1 - 2t_j/L$.
Q_n, C_n	Rational-kernel denominator and the integer normalization factor.
d_n	Least common multiple of $1, \dots, n$.
Π_n, γ	Available common divisor and its exponential rate, or a certified lower rate when specified.
λ, Θ	$\lambda = \log \mathrm{i} = \mathrm{i}\pi/2$, $\Theta = \pi/2$.
Y	Positive contour height after rescaling by n; not the polynomial indeterminate in a particular formula.
D_0, G_0	Decay and growth rates before division by the common divisor.
D, G	Usable rates after division, possibly weakened to conservative rational bounds.
Δ_k	Effective gap $D - (k-1)G$, required to be positive.
s, v_j	Common scale and fixed base weights, so $\nu_j = sv_j$.
V, S	Base multiplicity $\sum_j v_j$ and weighted base length $\sum_j v_j c_j$.
p, q	Arbitrary rational-approximation numerator and positive denominator.
θ, y	Nearby algebraic numbers $(p/q)^{1/k}$ and $\mathrm{i}\theta/2$.

The notation Y is conventional both for a contour height and a polynomial variable. In a computation it is safer to use different names. Similarly, the small finite field-independent degree k of a defining binomial must not be confused with the possibly large auxiliary degree m.

C.2 Deriving the exponent without suppressing the index choice

Let $\eta > 0$ be the common slack in the three asymptotic estimates, and set $\Delta_\eta = D - (k-1)G - k\eta > 0$. The resultant lower bound has the form

$$|U(y)| \geq (2^k q)^{-m} e^{-(k-1)(G+\eta)n}.$$

Choose a multiple of L so that $n \geq (m \log(2^k q) + \log 2)/\Delta_\eta$. The remainder $e^{-(D-\eta)n}$ is then at most half this lower bound. The divided difference contributes one more growth factor, giving

$$|\lambda - y| \geq \tfrac{1}{2}(2^k q)^{-m} e^{-k(G+\eta)n}.$$

Since rounding increases n by less than L, its effect is a positive constant independent of q. The power of q is

$$m + \frac{mk(G+\eta)}{\Delta_\eta} = \frac{m(D+G)}{\Delta_\eta}.$$

This calculation explains both the denominator and the numerator of the final bound. It also shows why omitting the divided difference, or forgetting a conjugate in the resultant, changes the exponent rather than merely the prefactor.

C.3 Short exercises with solutions or hints

1. Rational rescaling. Show that $\mu(a\xi/b) = \mu(\xi)$ for nonzero integers a, b. A rational approximation to one side gives one to the other after multiplying numerator and denominator by fixed integers. After reduction, the denominator changes by a factor bounded above and below independently of the approximation. Apply this to $\beta(3) = \pi^3/32$ and $\zeta(4) = \pi^4/90$. Notice that taking powers does not have this bounded-denominator property in both directions.

2. The elementary arctangent inequality. Differentiate $\arctan u - u + u^3/3$. Its derivative is $u^4/(1+u^2) \geq 0$, and the value at zero is zero. This proves the inequality used in the root-free estimate without any alternating-series restriction such as $u \leq 1$.

3. Why the exact saddle is not necessary. For fixed multiplicities, differentiate $\Phi(Y) = \sum_j \nu_j F_{c_j}(Y) - \Theta Y - m$. The second derivative is negative. Thus the unique critical point is a global maximum, but every $Y > 1/2$ still provides a valid lower decay rate. A rational trial height trades a small amount of sharpness for shorter numerical certificates.

4. The degree obstruction. Rewrite $E_k = m(D+G)/(D-(k-1)G)$ as $m(1+kG/\Delta_k)$. If $G \geq 0$, then $E_k \geq m$. In particular, no refinement of these same rates at fixed $m = 171$ can yield an exponent below 100 through this formula. A new transfer argument or a smaller auxiliary degree would be needed. This does not rule out a different construction.

5. A smaller true algebraic degree. Take $p/q = (a/b)^k$ with $a, b > 0$. Then $y = \mathrm{i}a/(2b)$ already belongs to $\mathbb{Q}(\mathrm{i})$. Its minimal polynomial over that field has degree one, even though the defining binomial has degree k. Explain why the minimal-polynomial resultant remains nonzero for a selected row, and why a product over all roots of the binomial need not be.

6. A common divisor and the gap. Keeping D_0, G_0 fixed, increase the usable divisor rate by $h > 0$. The new pair is $(D+h, G-h)$, so the gap increases by kh while $D+G$ is unchanged. This is the precise reason the same divisor has to be used in both the analytic remainder and the coefficient height.

7. What a finite numerical test can establish. Suppose $-n^{-1}\log|Q_n\beta(3) - P_n|$ is positive at each of the first one hundred indices. This does not prove that the integer linear forms tend to zero, that their asymptotic rate is positive, or that the nonproportionality condition needed for every denominator holds. State the additional asymptotic and determinant hypotheses before extracting an irrationality bound. The failed experiments in Appendix B should be read with exactly this distinction in mind.

References

[1] M. Mignotte, *Approximations rationnelles de π et quelques autres nombres*, Journées arithmétiques (Grenoble, 1973), Mémoires de la Société Mathématique de France **37** (1974), 121–132. doi:10.24033/msmf.139.

[2] M. Kawashima, *Linear independence of values of logarithms revisited*, arXiv:1904.01737 (2019). https://arxiv.org/abs/1904.01737.

[3] R. Marcovecchio, *Multiple Legendre polynomials in Diophantine approximation*, International Journal of Number Theory **10** (2014), 1829–1855; arXiv version 2512.12890. https://arxiv.org/abs/2512.12890.

[4] Yu. Nesterenko and M. Waldschmidt, *On the approximation of the values of exponential function and logarithm by algebraic numbers*, Mat. Zapiski **2** (1996), 23–42; English preprint arXiv:math/0002047 (2000). https://arxiv.org/abs/math/0002047.

[5] R. Marcovecchio and W. Zudilin, *Hypergeometric rational approximations to $\zeta(4)$*, Proceedings of the Edinburgh Mathematical Society **63** (2020), 374–397; preprint arXiv:1905.12579 (2019). doi:10.1017/S0013091519000427.

[6] F. Johansson, *Arb: Efficient Arbitrary-Precision Midpoint-Radius Interval Arithmetic*, arXiv:1611.02831. https://arxiv.org/abs/1611.02831.